impara
CREATIVE LABS
Learn. Create. Build. Beyond.

IoT & SMART AUTOMATION



3-MONTH INDUSTRY-SUPPORTED CAREER PROGRAM

Duration
3 Months / 12 Weeks

Mode
Technical Training +
Hands-on Learning + Project Development

LEARN IOT.
BUILD 3 PROJECTS.
CREATE SMART,
CONNECTED SOLUTIONS.

2 Mini Projects
+
1 Capstone Project

Complete the course with 3 hands-on projects and the skills to build connected, automated IoT solutions.

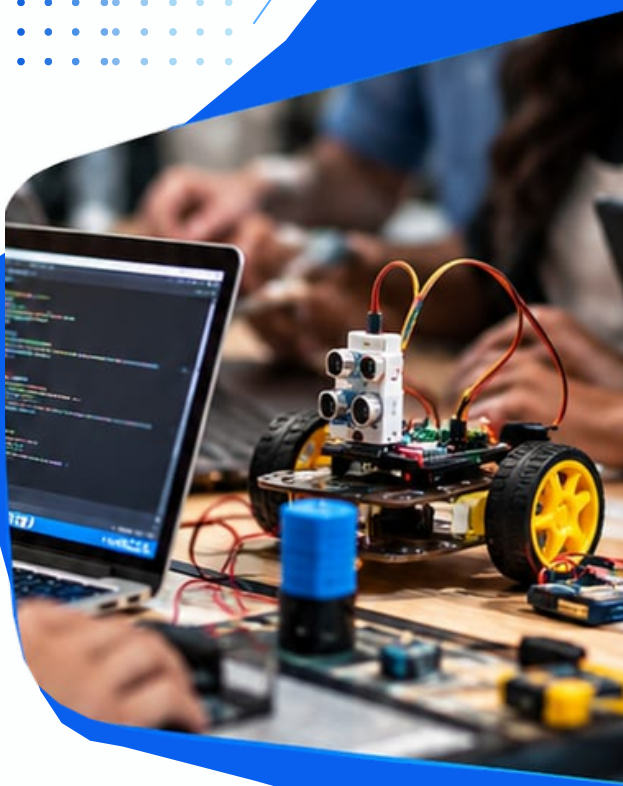
impara CREATIVE LABS
Learn. Create. Build. Beyond.

TECHKNOTS
One knot at a time

ABOUT

IMPARA

POWERED BY TECHKNOTS



Learn. Create. Build. Beyond.

Impara Creative Labs is a student-focused learning and career development platform that helps students turn technical knowledge into practical skills. Through hands-on training, project-based learning, mentorship and industry exposure, Impara helps students learn, build and prepare for real-world technology careers.

POWERED BY TECHKNOTS
Built on the technology, R&D and industry experience of TechKnots.

WHAT WE DO

Industry-Oriented Learning
Practical learning built around relevant skills, tools and technologies used in real-world development.

Hands-On Projects
Learn by building practical projects and applying concepts.

Industry Exposure
Understand real-world workflows, tools and project environments.

Career Preparation
Build technical skills, project experience and professional readiness.

OUR FOCUS AREAS

- IoT & Smart Automation
- Full Stack Development
- AI & Generative AI
- Data & Emerging Technologies

POWERED BY TECHKNOTS

- R&D & Technology
- Industry Experience
- Technology Innovation
- Student Ecosystem

Impara helps students build real-world skills, not just learn theory.

impara CREATIVE LABS
Learn. Create. Build. Beyond.

TECHKNOTS
One knot at a time

MEET THE FOUNDER & CEO

Researcher. Innovator. Mentor.

Impara was built on a simple belief: students learn best by building, experimenting and solving real-world problems.



NIXON E

Founder & CEO, Impara Creative Labs | Founder, TechKnots

Research Engineer | Technology Innovator | Educator

ABOUT NIXON E

Nixon E is a technology researcher and innovator with 7+ years of experience across AI, IoT, embedded systems, robotics and automation.

He founded TechKnots to develop practical technology solutions through R&D, innovation and project-based learning.

He has mentored 10,000+ engineering students through technology projects and innovation activities.

ACADEMIC BACKGROUND
B.E. EEE | M.Tech Power Electronics
M.E. Embedded Systems | MBA Scholar | DBA Scholar

7+ YEARS OF R&D EXPERIENCE
AI | IoT | Embedded Systems | Robotics | Automation

10,000+ STUDENTS MENTORED
Technology projects, innovation and R&D activities

IEEE MEMBER
Madras Section | 5+ years of professional participation

RESEARCH & INNOVATION FOCUS

- Artificial Intelligence
- IoT & Embedded Systems
- Smart Agriculture
- Biomedical Technology
- Robotics & Autonomous Systems
- Industrial IoT & Digital Twin
- Smart Infrastructure & Safety

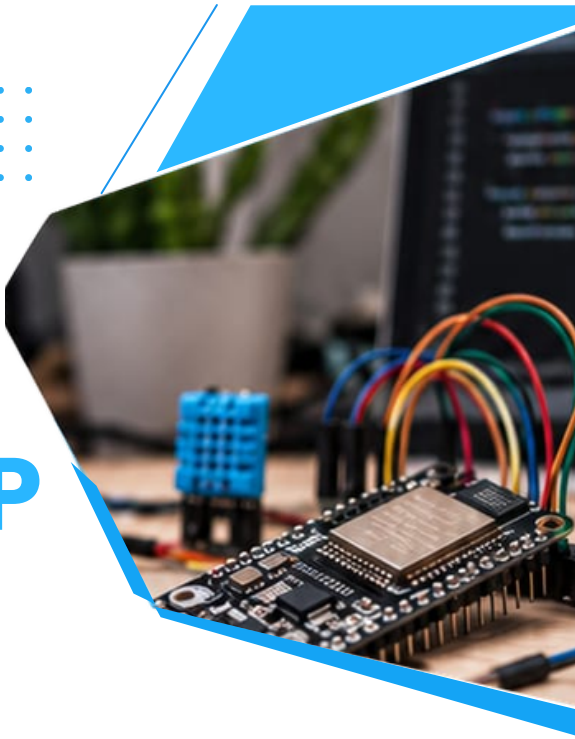
We don't just teach technology. We help students build with it.

OUR VISION
Make practical technology education accessible.

OUR FOCUS
Hands-on learning, real-world projects and industry exposure.

OUR MISSION
Turn technical knowledge into practical skills.

impara
CREATIVE LABS
Learn. Create. Build. Beyond.



12-WEEK COURSE ROADMAP

A structured learning journey designed to take you from fundamentals to real-world, project-based IoT solutions.

WEEKS 1 - 3

IoT FUNDAMENTALS

- Introduction to IoT, IIoT & AIoT
- Types and stages of IoT
- Sensors and inputs
- Microcontrollers & microprocessors
- Actuators and outputs
- Basic IoT architecture

01

WEEKS 4 - 6

EMBEDDED SYSTEMS

- Arduino Uno, Nano & Mega
- ESP8266 & ESP32
- Pin configuration
- Sensors, controllers & actuators
- Arduino IDE
- VS Code + PlatformIO
- Circuit design & simulation

02

WEEKS 7 - 9

CONNECTIVITY & COMMUNICATION

- IoT communication protocols
- MQTT
- HTTP
- Webhooks
- CoAP
- AMQP
- SMTP
- SNMP
- RFID & NFC
- IPv4, IPv6 & MAC
- IEEE standards
- Local IP & mDNS

03

WEEKS 10 - 12

CLOUD, AUTOMATION & PROJECT DEVELOPMENT

- Blynk
- AWS
- ThingSpeak
- IFTTT
- ESP32-CAM
- Computer Vision applications
- Basic DIP
- System integration
- Project development

04

LEARNING JOURNEY

LEARN
Build a strong foundation in IoT and embedded systems.

BUILD
Work hands-on with tools, hardware and real-world concepts.

CONNECT
Integrate devices, explore communication protocols and cloud platforms.

AUTOMATE
Develop smart, connected solutions through real projects.

impara
CREATIVE LABS
Learn. Create. Build. Beyond.



TECHNOLOGIES & TOOLS

Work with industry-relevant hardware, software and platforms to build real-world IoT solutions.

01 DEVELOPMENT BOARDS

Work with popular microcontroller and IoT development boards.


Arduino Uno


Arduino Nano


Arduino Mega


ESP8266


ESP32


ESP32-CAM

02 SENSORS & ACTUATORS

Learn to interface and work with real-world input and output devices.

- Sensors & Input Devices
- Actuators & Output Devices
- Microcontrollers & Processors
- Pin Configuration & Interfacing

03 DEVELOPMENT & CODING

Use industry-standard development environments and coding tools.

- Arduino IDE
- VS Code
- PlatformIO

04 SIMULATION & CIRCUIT DESIGN

Design, simulate and test your projects before building them.

- Tinkercad
- Wokwi
- Fritzing

05 CLOUD & IoT PLATFORMS

Build and deploy connected IoT solutions using cloud platforms.

- Blynk
- AWS
- ThingSpeak

06 CONNECTIVITY & AUTOMATION

Work with various communication protocols and automation platforms.

- MQTT
- HTTP
- Webhooks
- CoAP
- AMQP
- SMTP
- SNMP
- IFTTT

07 IDENTIFICATION & NETWORKING

Explore identification technologies and network communication standards used in IoT.

- RFID
- NFC
- IPv4
- IPv6
- MAC
- IEEE Standards

impara
CREATIVE LABS
Learn. Create. Build. Beyond.



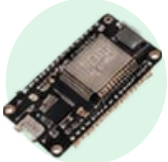
SKILLS YOU'LL BUILD

Gain practical, industry-relevant skills to design, develop and deploy real-world IoT solutions.

01 IoT SYSTEM DESIGN

- Understand IoT, IIoT & AIoT architectures
- Design basic IoT systems
- Work with sensors, controllers and actuators
- Configure device pins and interfaces

02 EMBEDDED DEVELOPMENT



- Program Arduino boards
- Work with ESP8266 and ESP32
- Interface sensors and actuators
- Build and test embedded systems

03 CONNECTIVITY & PROTOCOLS

- Connect IoT devices to networks
- Work with MQTT, HTTP, Webhooks and other protocols
- Understand IP addressing and networking
- Implement device-to-device communication

04 CLOUD & SMART AUTOMATION

- Connect devices to Blynk, AWS and ThingSpeak
- Build automated workflows with IFTTT
- Work with local IP and mDNS
- Develop connected IoT applications

05 COMPUTER VISION & INTEGRATION

- Work with ESP32-CAM
- Explore basic Computer Vision
- Apply basic DIP
- Integrate hardware, software and connectivity into complete systems

FROM CONCEPT TO REAL-WORLD SOLUTIONS

FROM **DEVICE**
Work with real hardware and sensors.

TO **DATA**
Collect, process and transmit data.

TO **AUTOMATION**
Build smart, connected IoT solutions.



impara
CREATIVE LABS
Learn. Create. Build. Beyond.

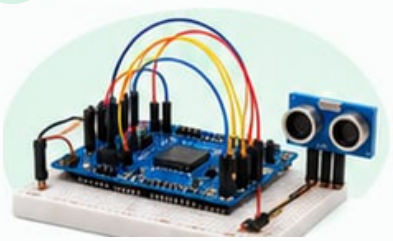


PROJECT-BASED LEARNING

2 MINI PROJECTS + 1 CAPSTONE PROJECT

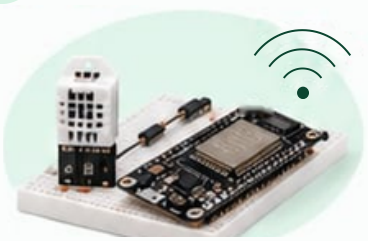
A progressive learning experience to help you apply concepts, solve real problems and build a complete IoT solution.

01 MINI PROJECT 1



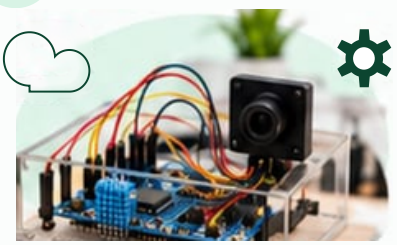
- Apply the fundamentals of sensors, microcontrollers, inputs and outputs.
- Build and test a basic IoT system.
- Gain hands-on experience with hardware and coding.

02 MINI PROJECT 2



- Build a connected IoT system using communication, cloud or automation concepts.
- Implement real-time data collection and transmission.
- Explore integration with platforms and protocols.

03 CAPSTONE PROJECT



- Integrate hardware, connectivity, software and automation into a complete IoT solution.
- Solve a real-world problem with an end-to-end system.
- Build a portfolio-ready project to showcase your skills.

WHAT YOU'LL PRACTICE

Sensors & Inputs → Microcontrollers & Processors → Communication & Protocols → Cloud Platforms → Automation & Integration → Outputs & Real-World

3 HANDS-ON PROJECTS

1 COMPLETE IoT DEVELOPMENT JOURNEY

From concept to a working, real-world IoT solution.



impara
CREATIVE LABS
Learn. Create. Build. Beyond.



BEYOND THE CLASSROOM

Go beyond technical learning to help you build a strong portfolio, gain industry exposure and get career-ready.

01

PORTFOLIO-READY PROJECTS

Build 3 real-world projects (2 mini projects + 1 capstone) to showcase your skills.

02

INDUSTRY EXPOSURE

Learn through real-world use cases, industry-relevant tools and guidance from experienced mentors.

03

EXPERT MENTORSHIP

Get guidance from industry professionals throughout your learning journey.

04

CERTIFICATION

Receive a course completion certificate to validate your skills and project experience.

05

LINKEDIN PROFILE SUPPORT

Get help in optimising your LinkedIn profile to showcase your skills, projects and achievements effectively.

06

MOCK INTERVIEWS

Practice with mock interviews to build confidence and prepare for real opportunities.

CAREER OPPORTUNITIES

This program prepares you for exciting opportunities in the IoT and related domains.

- IoT Developer
- Embedded Systems Developer
- IoT Solutions Engineer
- Embedded IoT Engineer
- Smart Automation Engineer
- IoT Project / Technical Associate

Explore opportunities across startups, product companies and research organisations.

YOUR JOURNEY WITH IMPARA

LEARN
Industry-relevant skills with hands-on training

BUILD
3 real-world projects

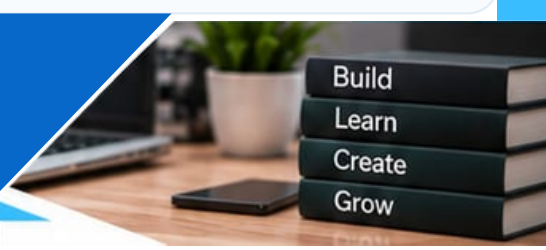
SHOWCASE
Create a strong portfolio and LinkedIn profile

PREPARE
With mock interviews and expert guidance

GROW
Be ready for opportunities in the industry

SKILLS TODAY. OPPORTUNITIES TOMORROW.

Gain the practical experience, portfolio and confidence to take the next step in your IoT journey.



impara
CREATIVE LABS
Learn. Create. Build. Beyond.



TAKE THE NEXT STEP IN YOUR IoT JOURNEY

Learn, build and gain real-world experience with Impara. Join our 3-Month IoT Program and develop the skills to create smart, connected solutions.

PROGRAM DURATION
3 Months
(12 Weeks)

MODE
Online / Offline
(As per batch)

NEXT BATCH
September 2026

CERTIFICATION
Course completion certificate from Impara

READY TO GET STARTED?

Scan the QR code or fill out the form to register your interest. Our team will contact you with the detailed program information, batch schedule and next steps.

SCAN TO REGISTER

OR

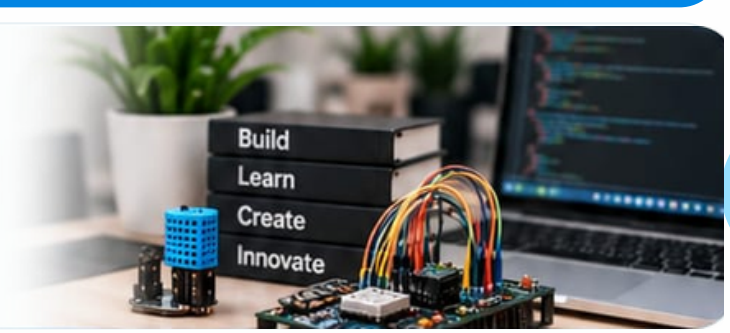
Reply on WhatsApp for more details

Email us at **info@impara.in**

Call us **+91 98765 43210**

LET'S BUILD A SMARTER TOMORROW.

Gain practical skills, build real-world projects and be a part of the next generation of innovators with Impara.



impara
CREATIVE LABS
Learn. Create. Build. Beyond.

Chennai, India
Working with students and institutions across India.

50+
Academic Collaborations

100+
Industry Projects

AICTE
NIRF Approved Internships